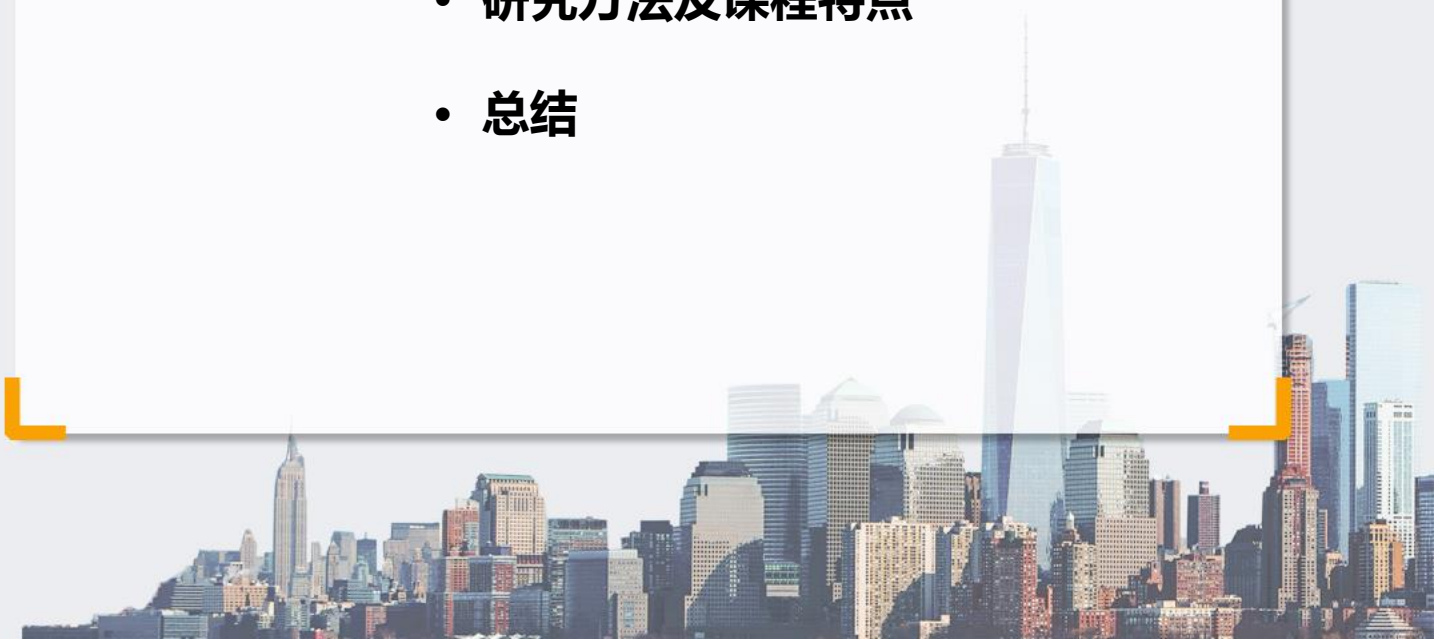
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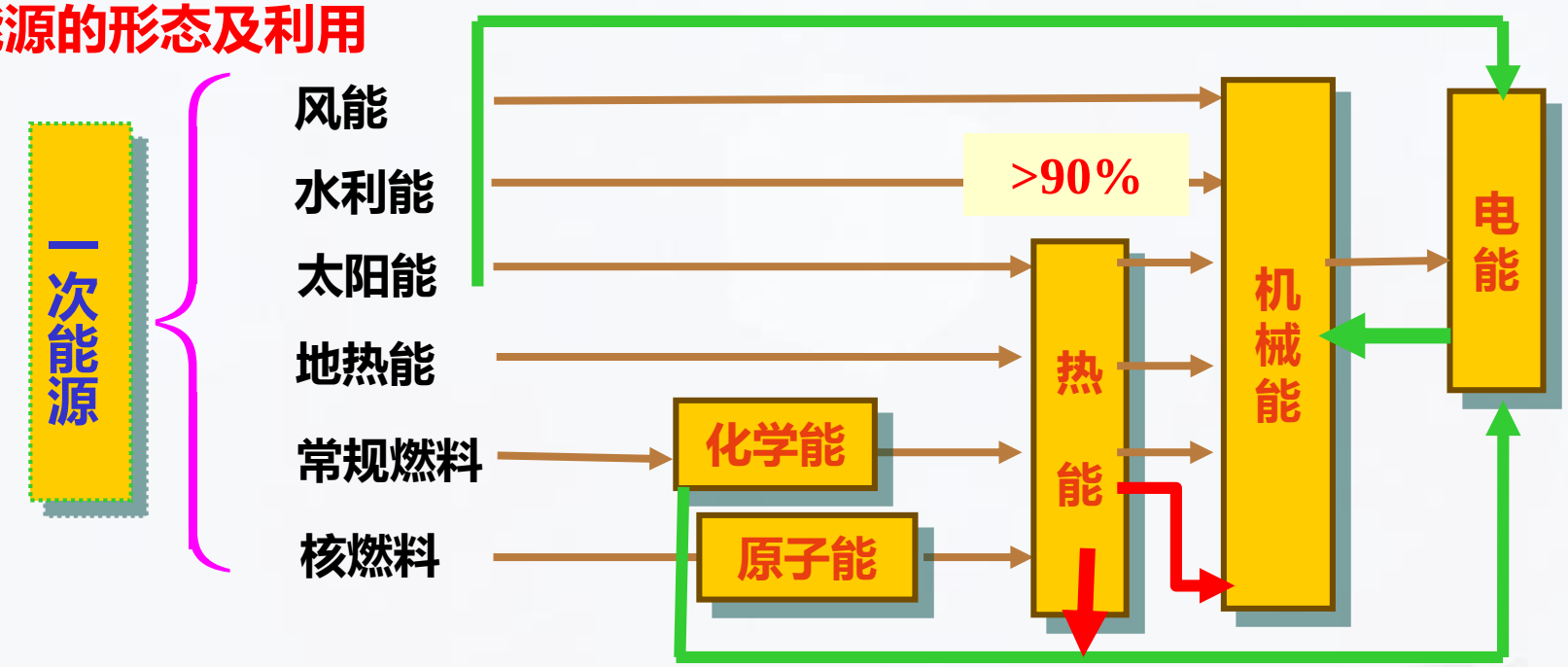
- 工程热力学
- 主要内容
- 研究方法及课程特点
- 总结



工程热力学是能源科学的一门基础理论课。

1.能源： 为人类生产与日常生活提供各种能量和动力的物质资源。

2.能源的形态及利用



热能利用的两种方式

直接利用（热利用）：　热能直接用于加热物体

间接利用（能量转化）：　热能 → 机械能　>80%

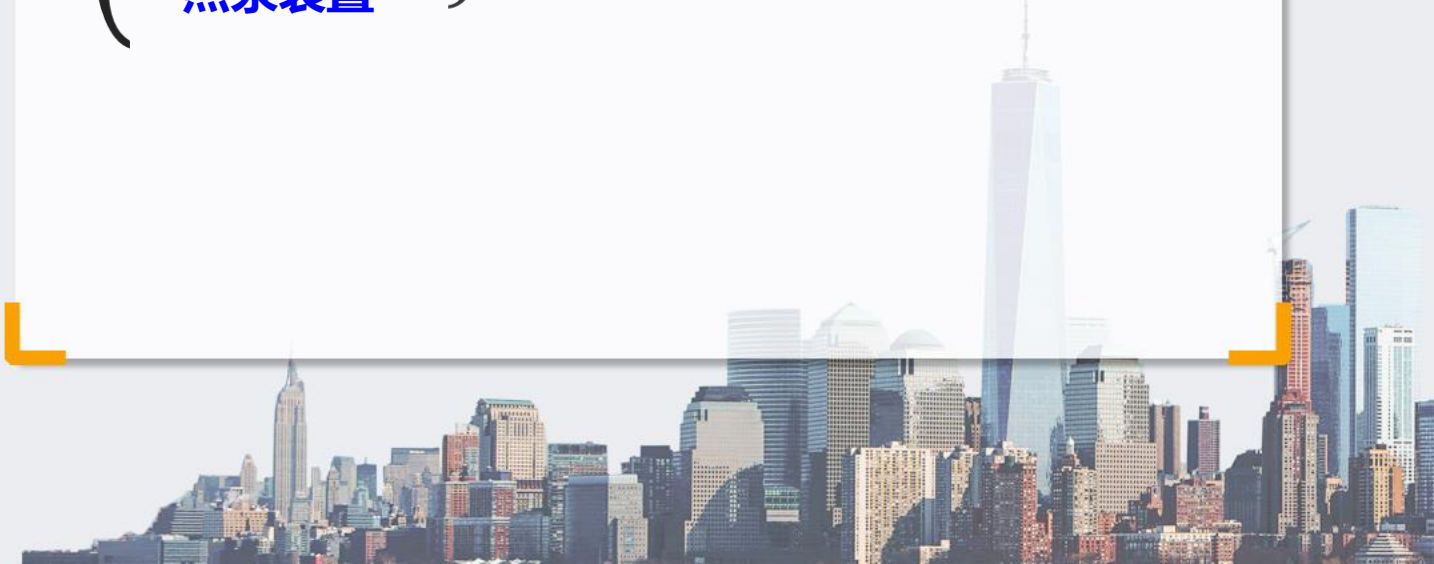
3. 工程热力学的研究对象

研究热能和机械能之间的相互转化规律；并探讨提高转换效率的途径。

建立起能量的开发与利用以及节能的理论基础

热能转换装置：

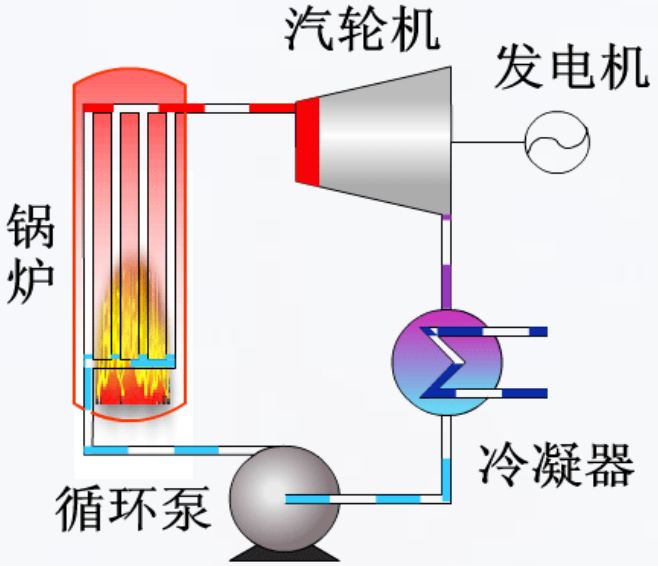
- **动力装置**：热能 → 机械能
- **制冷装置**、**热泵装置**：机械能 → 热能

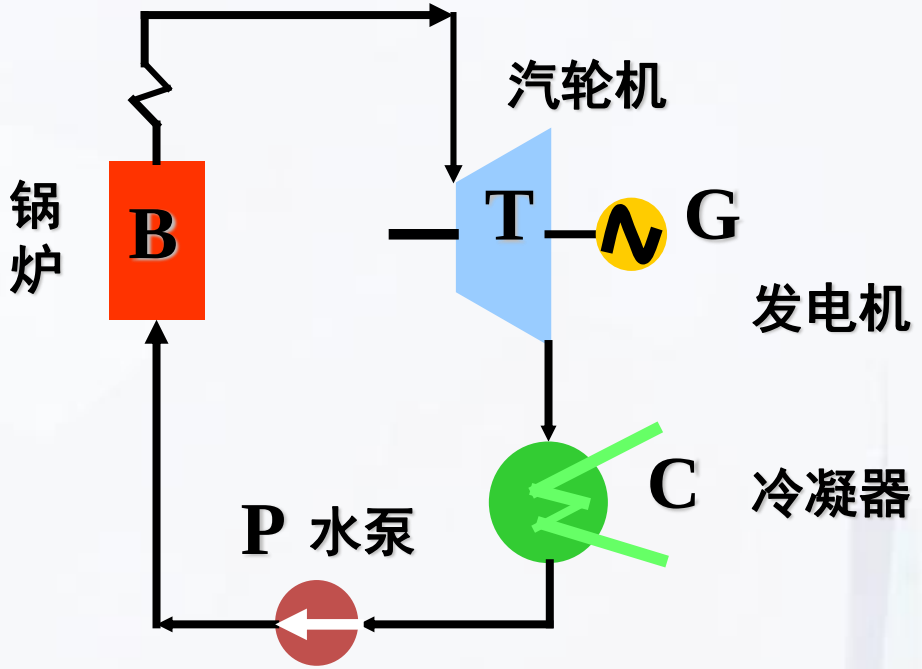


例1 蒸汽动力装置



简单蒸汽动力装置循环-朗肯循环(Rankine cycle)

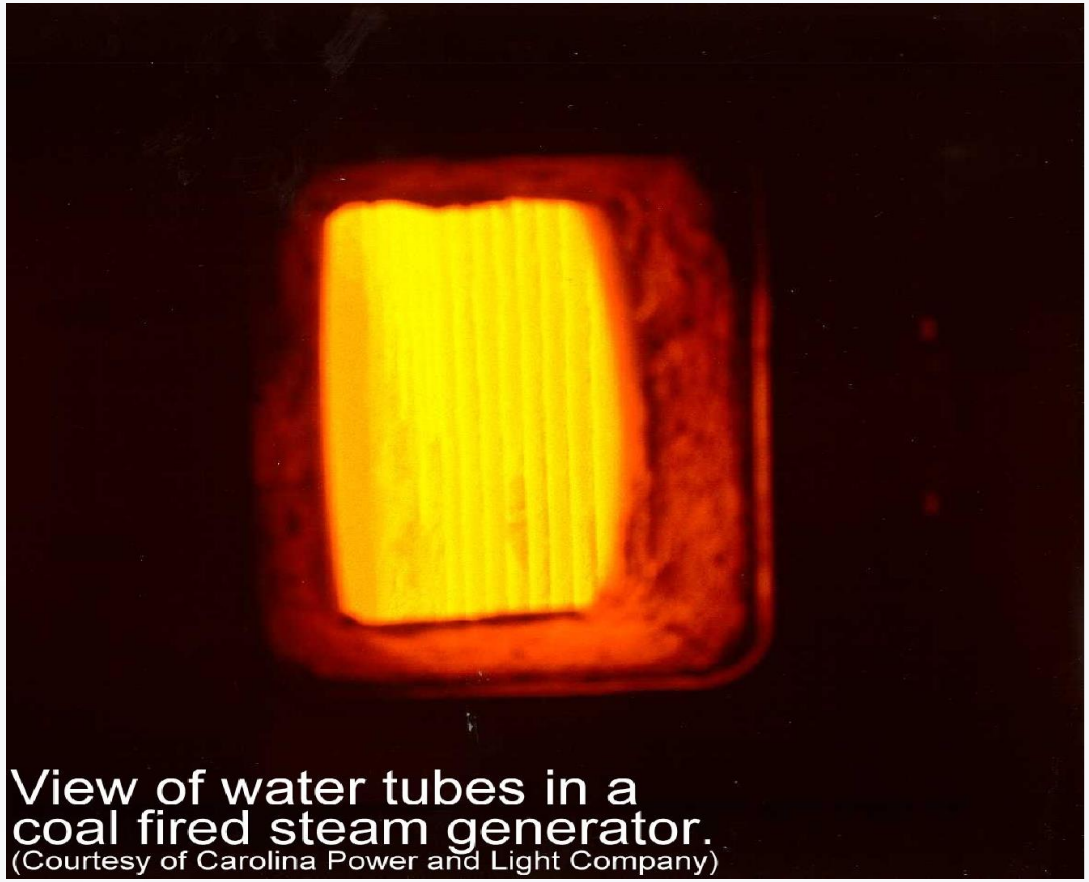




锅炉



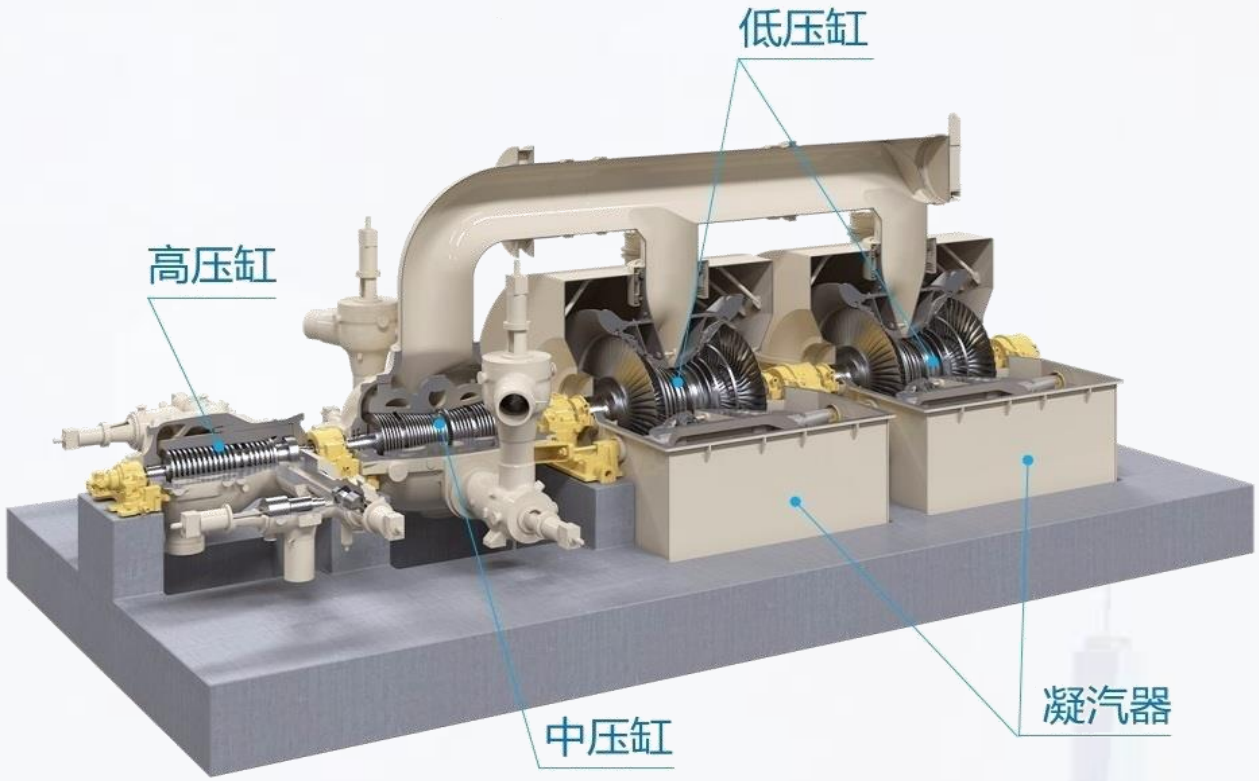
Outer surface of a steam generator at a coal fired steam steam power plant. (Courtesy of Carolina Power and Light Company)



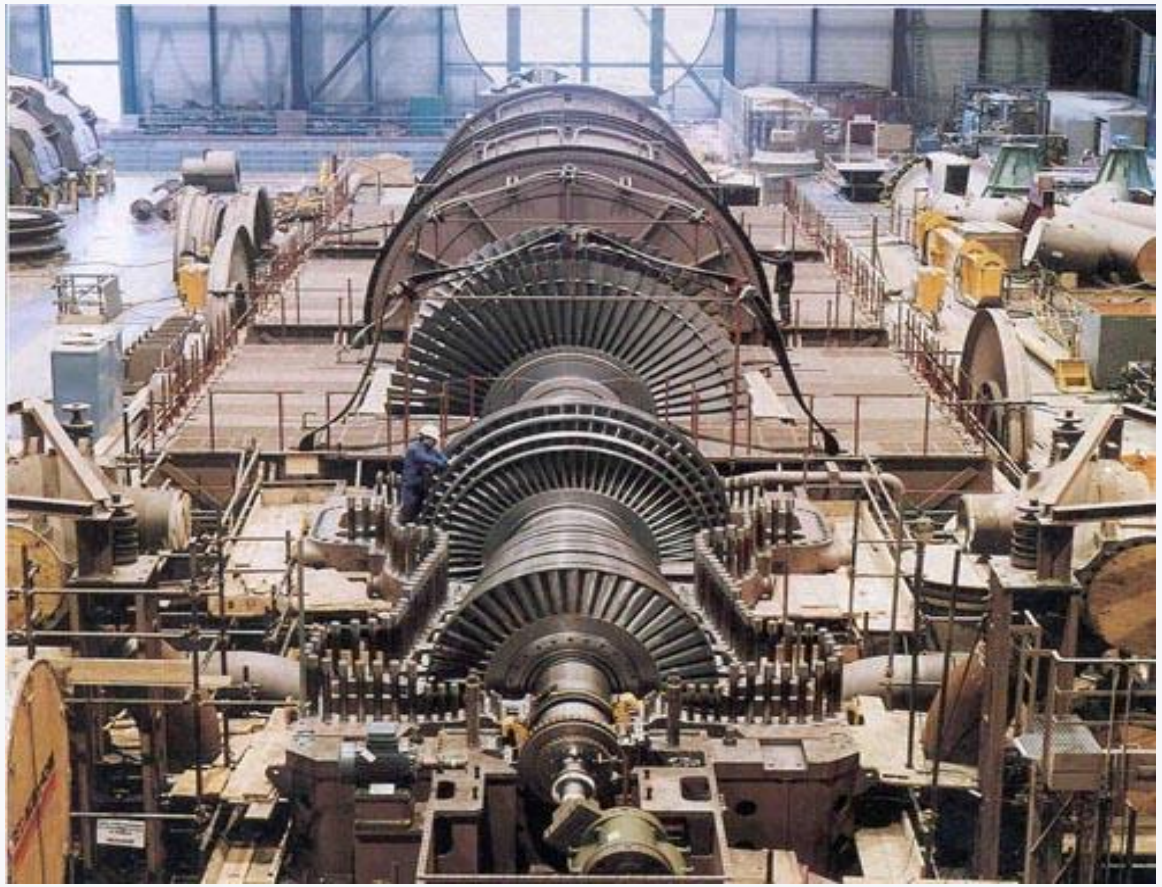
View of water tubes in a coal fired steam generator.
(Courtesy of Carolina Power and Light Company)

汽轮机

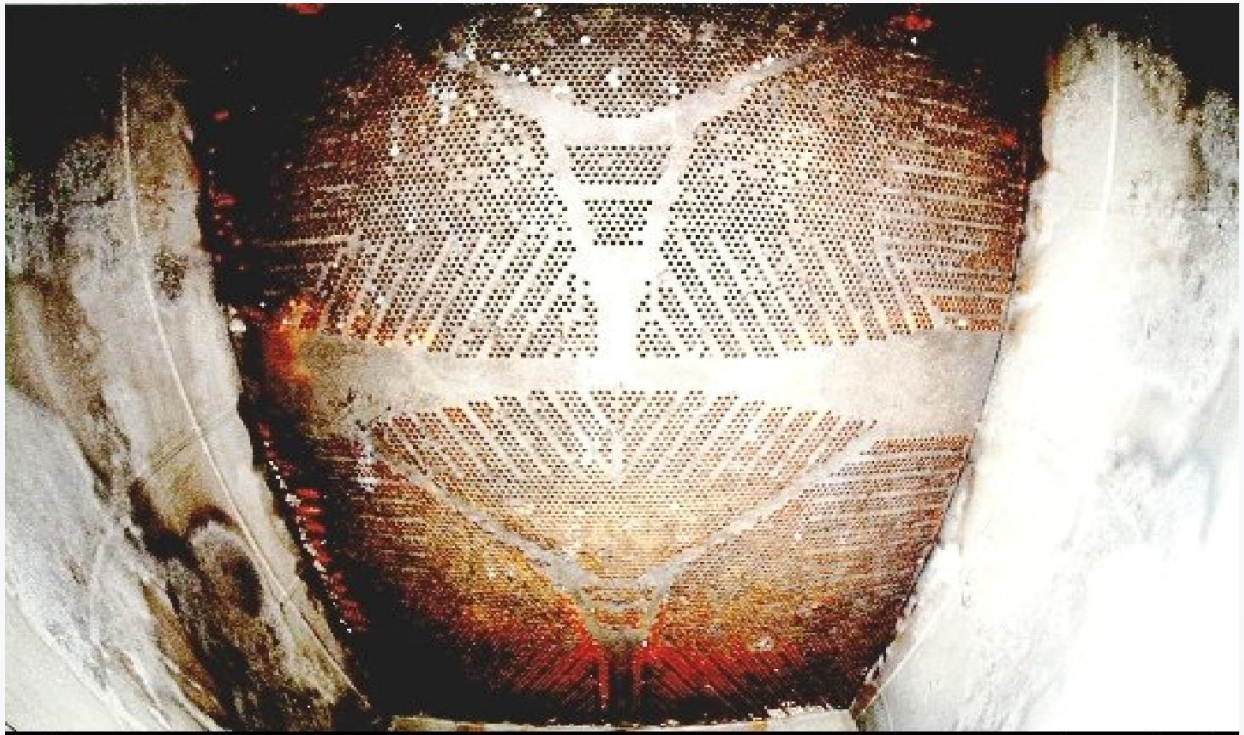
高温高压蒸汽依次流经高压缸、中压缸、低压缸，推动透平转动，同轴驱动发电机发电，乏汽排入凝汽器。



汽轮机



冷凝器



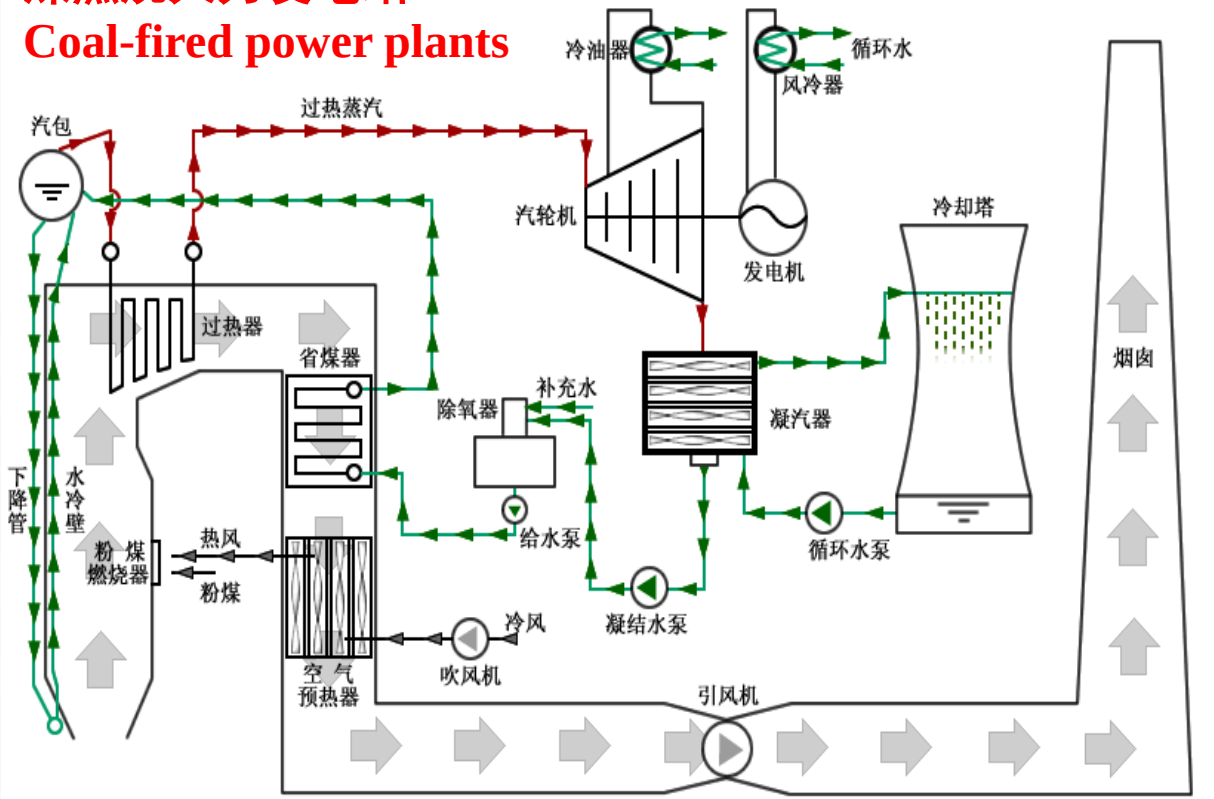
View of the entrance to the cooling water tub
tubes in a steam condenser.
(Courtesy of Carolina Power and Light Company)

冷却塔

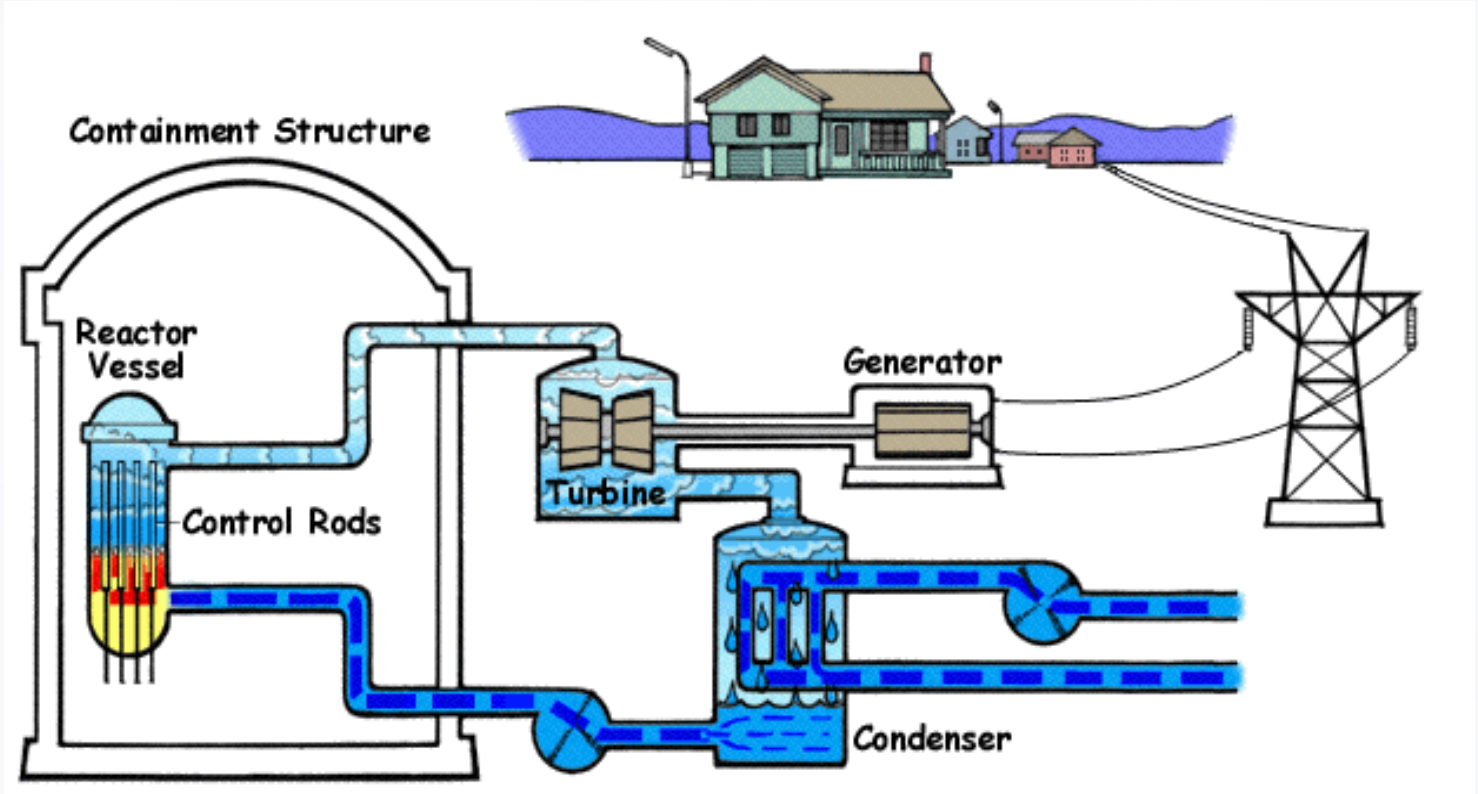


Aerial view of a nuclear steam power plant with cooling tower. (Courtesy of Carolina Power and Light Company)

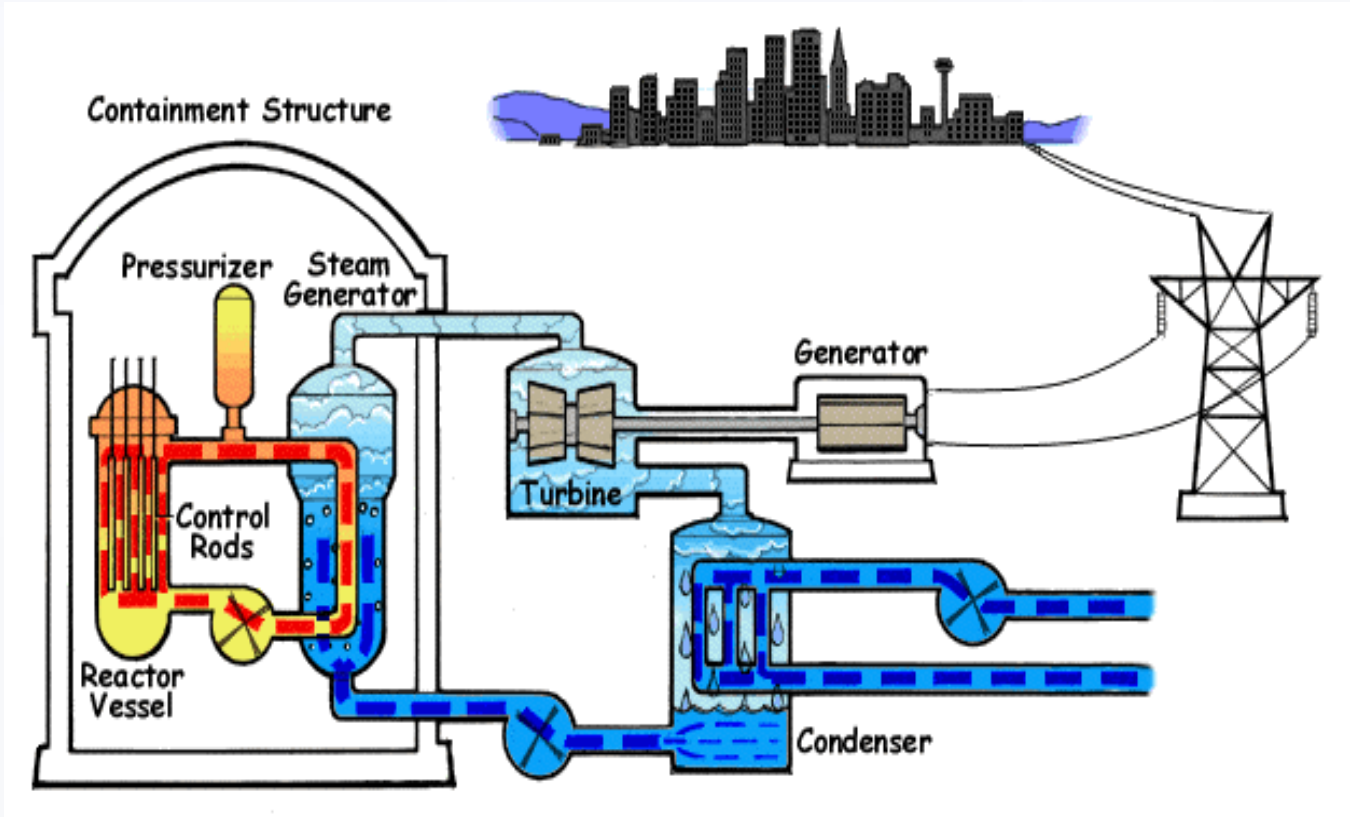
煤燃烧火力发电站
Coal-fired power plants



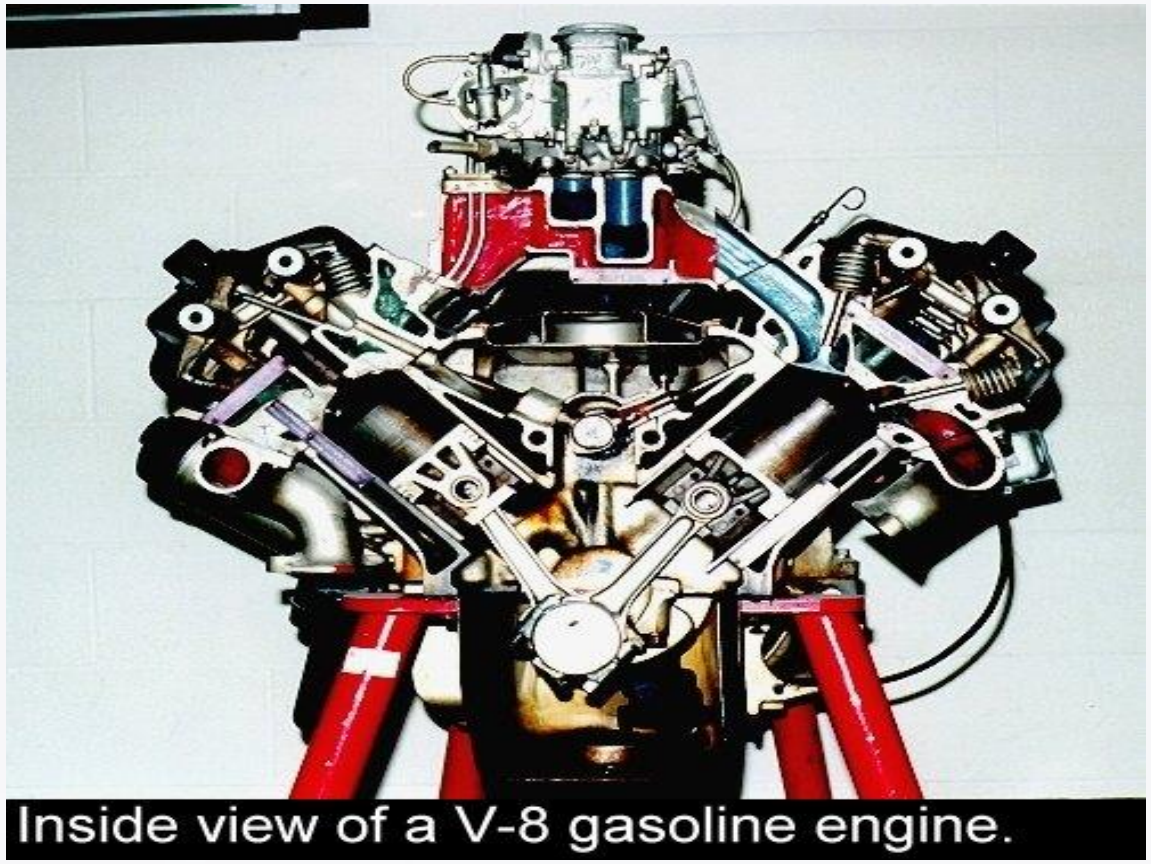
核电站示意图-沸水堆（Nuclear-Boiling Water Reactor）



核电站示意图-压水堆（Nuclear-Pressurized Water Reactors）



内燃机



Inside view of a V-8 gasoline engine.

船





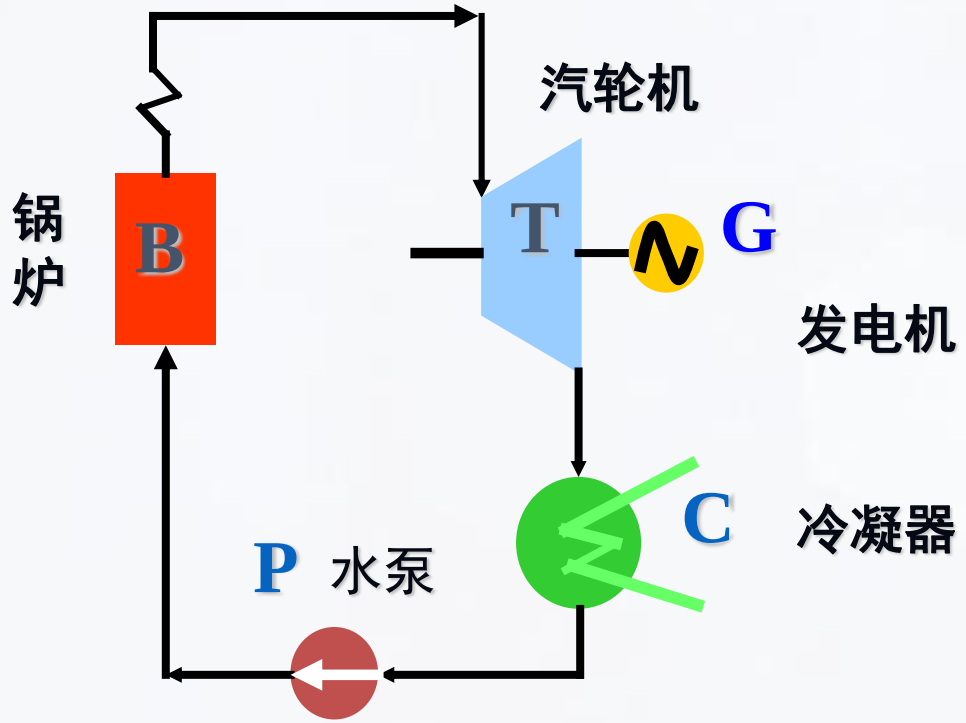
飞机



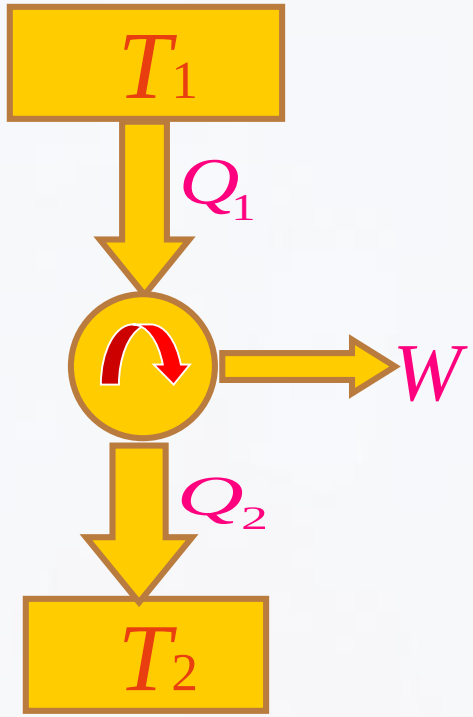




简单蒸汽动力装置循环-朗肯循环(Rankine cycle)



流程图



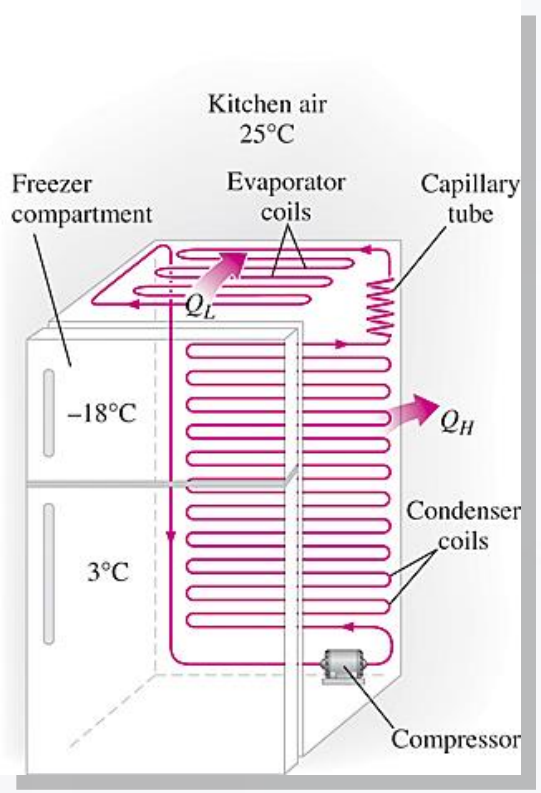
热力学原理图

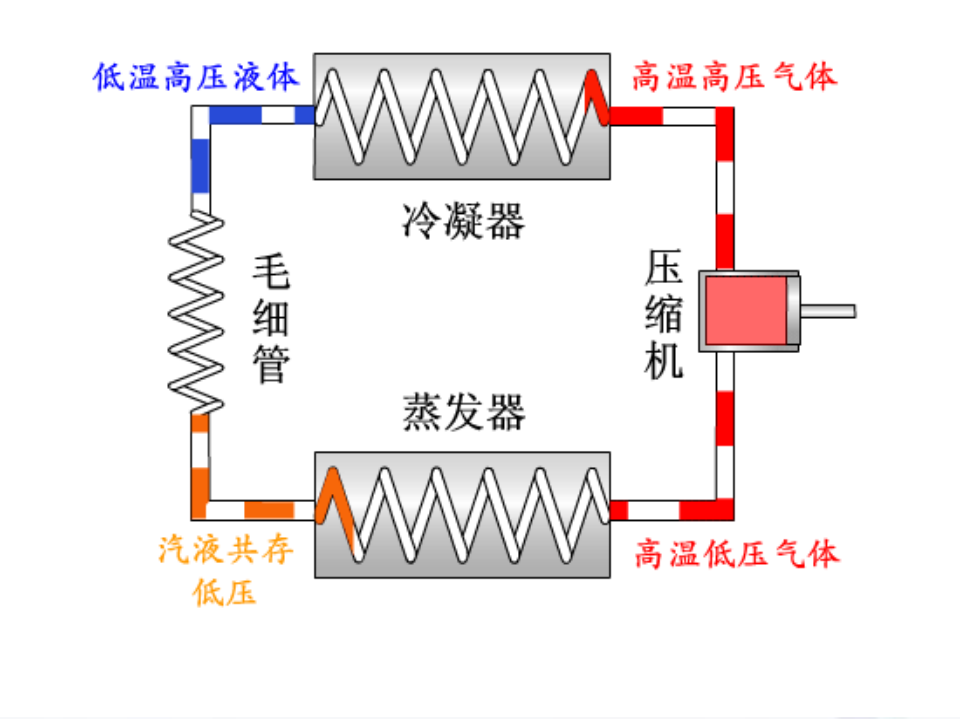
例2　制冷装置



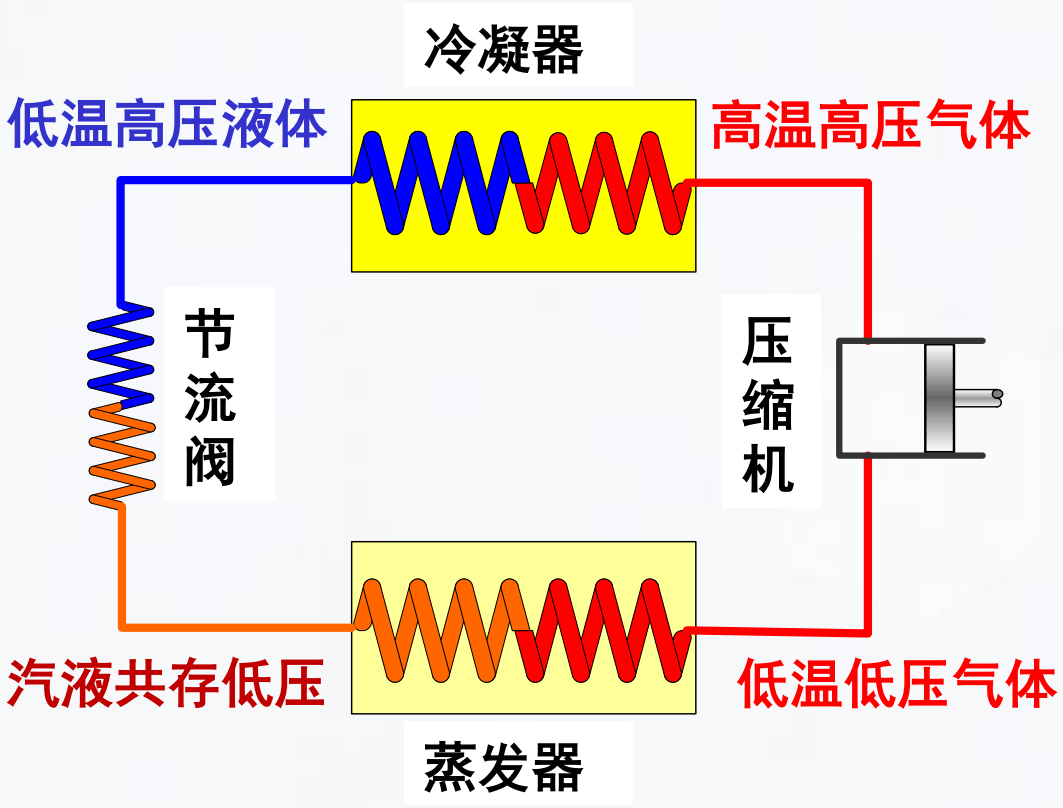


例2　制冷装置

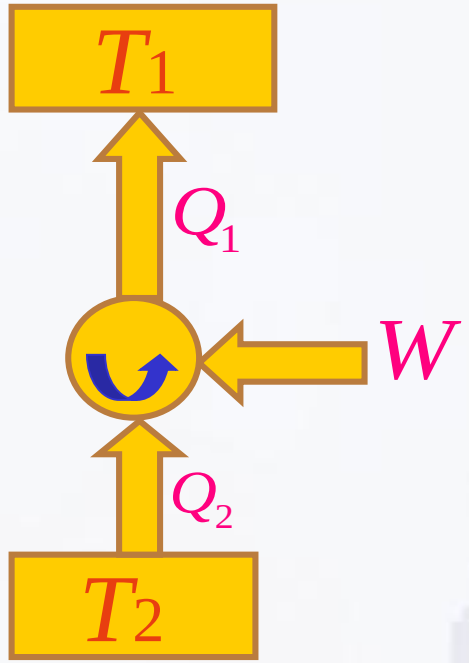

Kitchen air
25°C
Freezer
compartment
Evaporator
coils
Capillary
tube
Q_L
−18°C
Q_H
Condenser
coils
3°C
Compressor


低温高压液体
高温高压气体
冷凝器
毛
细
管
压
缩
机
蒸发器
汽液共存
低压
高温低压气体

流程图

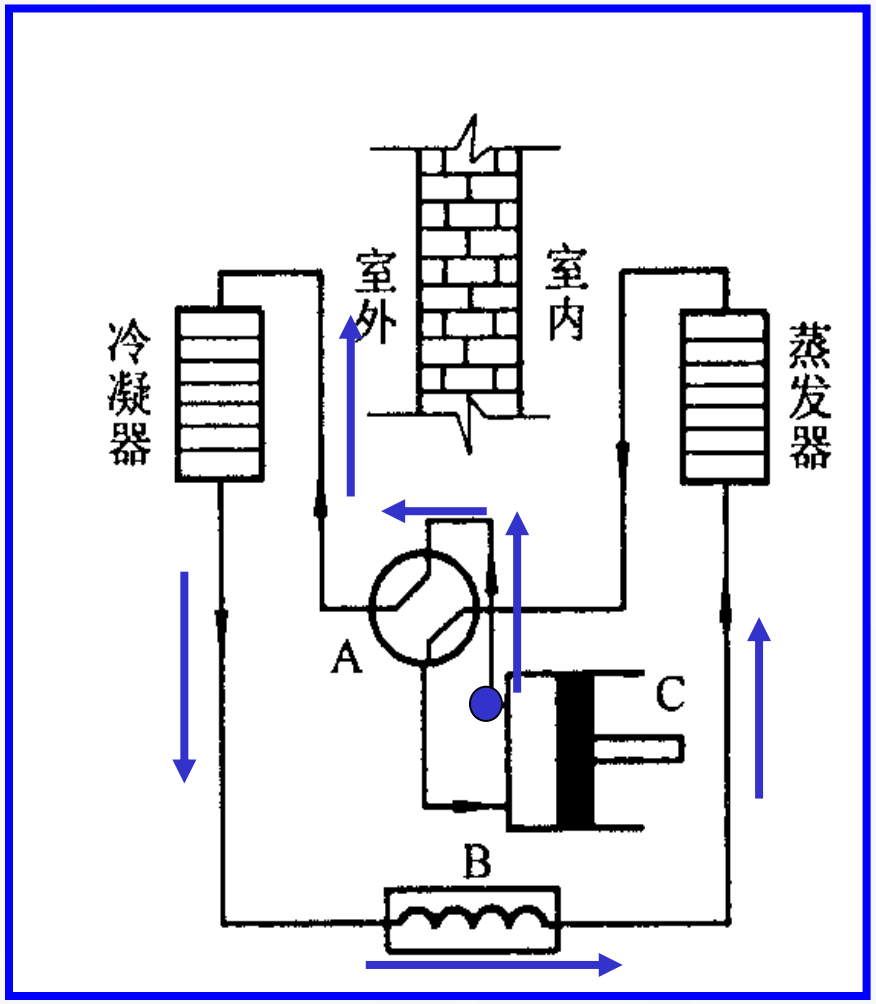


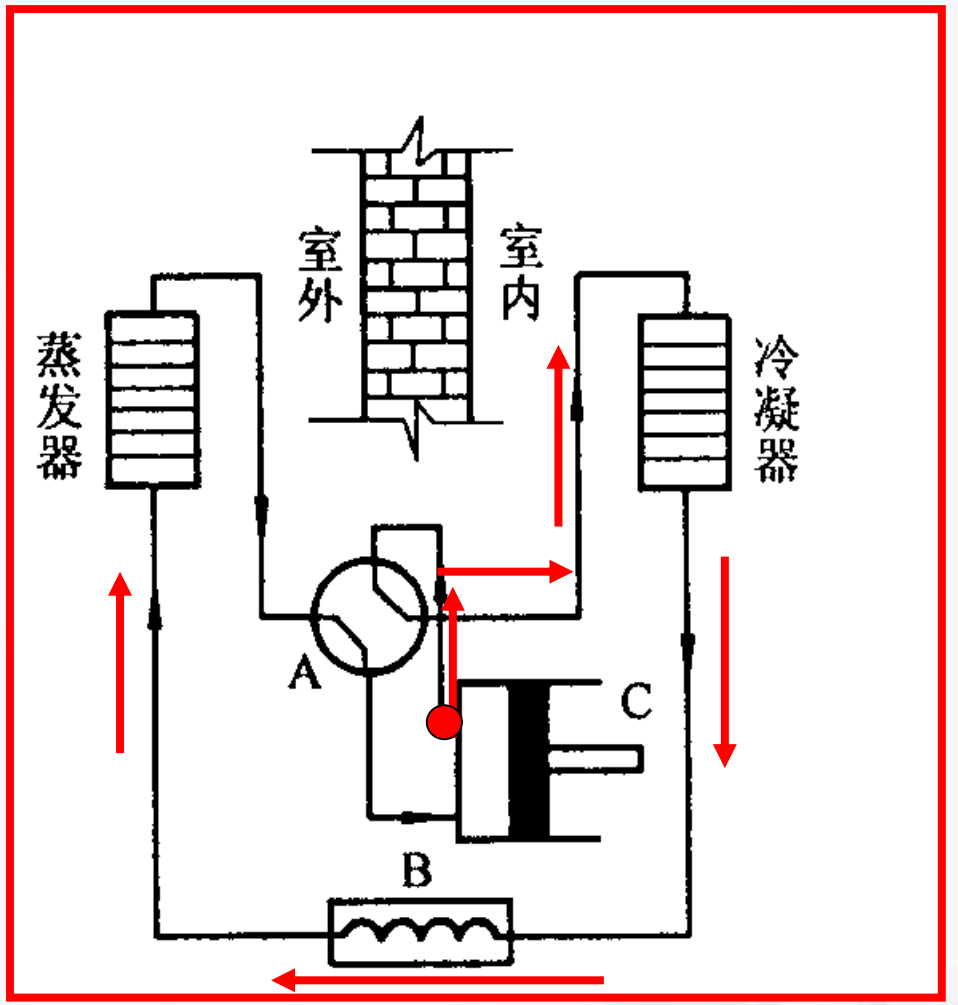
流程图



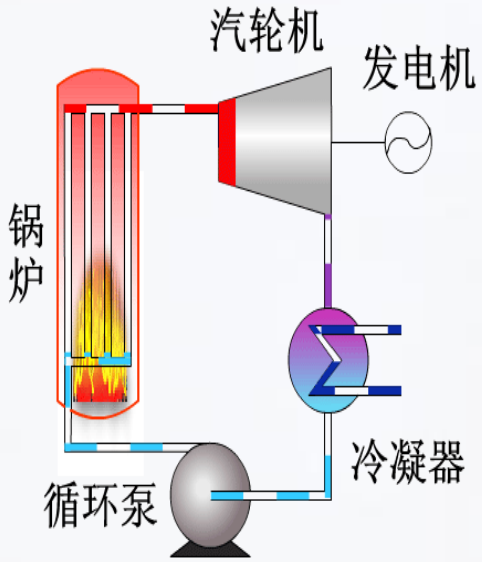
热力学原理图

夏季制冷循环

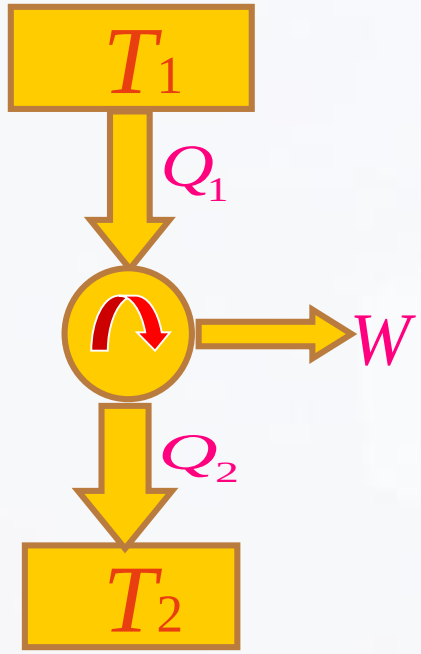




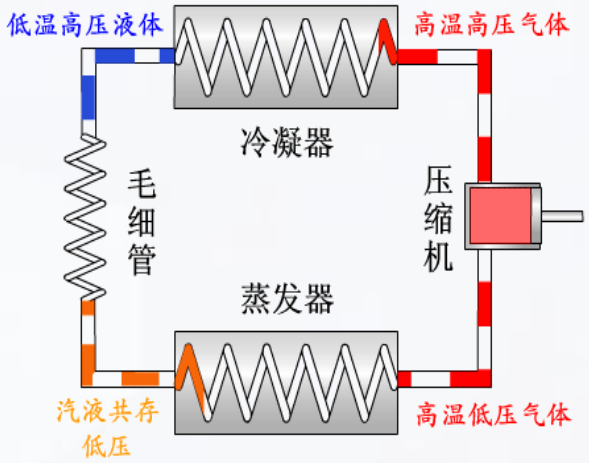
冬季制热循环



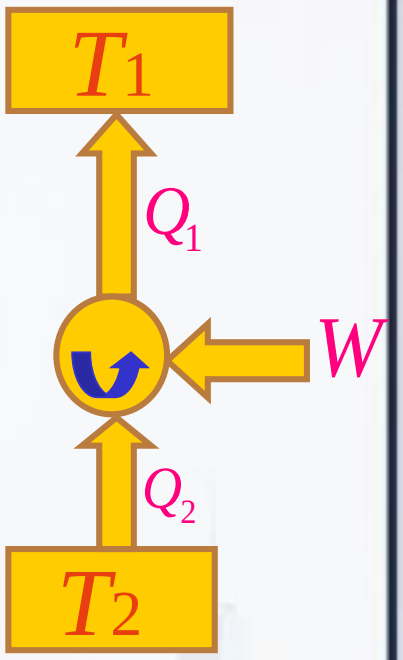
流程图



热力学原理图



流程图



热力学原理图

关心热能与机械能相互转换过程中一些规律性的、共同的内容。

主要内容	细 化
基本概念、基本定律	热力学第一定律 热力学第二定律
工质的热力性质	理想气体 实际气体
热力过程和热力循环	热力过程 热力循环
化学热力学	

1. 研究方法

- **微观方法：** 物质由微粒组成，着眼于微粒行为统计方法，统计热力学。
- **宏观方法：** 将工质看成整体，着眼于整体行为，唯象方法，经典热力学。
  classical, macroscopic



2. 课程特点

- 概念多，术语多
- 公式多，计算多
- 工程背景与科学抽象多
- 系统性强

辅助环节

现场参观	增强感性认识
实验	加强对理论的理解
习题课	培养灵活运用理论知识的能力
计算机	用现代计算方法解题

总 结

- 掌握热力学的基本原理、物理概念的实质（通过大量实际案例、例题、图、实践等帮助理解）；
- 通过与工程实际密切相关问题案例的分析和解决，初步学会理论如何应用于实际；
- 培养解决实际问题的能力、终身学习和工作的本领；
- To bridge the gap between knowledge and the confidence to properly apply knowledge.

Thermodynamics is a fascinating subject that deals with energy.
It has a broad application area ranging from microscopic organisms and household appliances, to vehicles, power generation systems, and even philosophy.